

Final Paper

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Introduction

The University of California, Santa Barbara (UCSB) North Campus Open Space (NCOS) is a 136-acre restoration site located northwest of the main UCSB campus in Goleta, California. Prior to restoration, the site was occupied by Ocean Meadows Golf Course, the construction of which deposited fill soil that eliminated approximately half of the site's original water-holding capacity. This hydrological disruption degraded the natural wetland habitat and increased flood risk to surrounding homes. Following restoration efforts led by UCSB's Cheadle Center

for Biodiversity and Ecological Restoration, the site received major recognition in September 2021 when FEMA issued a Letter of Map Revision (LOMR), formally removing the Flood Hazard Zone designation from nearby properties (Rickard 2024).

The restored site now supports a diverse array of native species across its varying wetland habitats, including perennial grassland, salt marsh, freshwater creek, and restored vernal pools. Among the key factors differentiating these habitats is salinity, which varies considerably across the site. Clark (2016) documented this variation in an early post-restoration water quality study, measuring salinity gradients alongside dissolved oxygen and nutrient levels to characterize overall wetland ecosystem health. Salinity levels at Venoco Bridge in particular were found to vary substantially in response to tidal influxes, rainfall, and freshwater inputs, with the greatest variation occurring around breach events connecting the slough to both freshwater and ocean sources.

This study investigates how changes in precipitation influence salinity levels across various sites at NCOS. It also explores how variables such as water elevation and temperature impact salinity. To explore this relationship, this study used the following data frames: “NOAA weather data”, “NCOS_YSI_Water_Quality_Monitoring”, and “YSI_Data_Begin,” to create visualizations to understand the broader environmental implications of this relationship. One application used from these data visualizations was observing changes in vegetation and how they relate to salinity, specifically through the elective map. Salinity is important to explore as it is known to have a strong impact on the biodiversity and species concentration within an ecosystem (Watt, García-Berthou, and Vilar 2006). For example, salinity is known to have an impact on waterfowl breeding, as waterfowl typically like to breed in freshwater habitats, but exist in more brackish water after their breeding season. NCOS’s unique wetland ecosystem allows for both freshwater habitats and brackish water habitats, making it a perfect environment for healthy abundance of waterfowl. With the restored status of NCOS, tracking salinity and waterfowl together can depict the relationship between waterfowl abundance and salinity. Salinity is an important proxy in measuring the restoration and health of species at NCOS, and thus is extremely important for seeing the success of the project (schacter2021?). This is because salinity has also been shown to potentially have a greater impact on ecosystems and its biota than other flows (Huckelbridge et al. 2010).

Through the following research questions, this study also explores the extent in which each of the factors affect salinity: How does salinity change with precipitation? How does salinity change with temperature? How does salinity change with water surface elevation?

Methods

The data used in this analysis were collected through differing methods and by differing parties. Understanding the temporal and spatial coverage of these datasets is important for providing context about the exact location of our information, as well as for the proper summarization of variables in subsequent direct comparison.

Measurements of the variables salinity (ppt) and water temperature ($^{\circ}\text{C}$), as well as conductivity and dissolved oxygen (not used for this analysis) were taken weekly by Cheadle Center student researchers at one foot depth intervals using a portable YSI Pro2030 at the three bridges that cross the upper slough: the Marsh trail bridges over the Phelps Creek outlet and “Dilling’s Link” across the east channel, and the Venoco access road bridge. From the bridges, the sensor is lowered to the water and data is recorded at the surface and at each foot of depth down to the bottom, or at each “elevation code”. Taking observations at each elevation code allows for the aspect of stratification to be taken into consideration. Or, in other words, the variation of these water parameters at different depths are important intricacies to understand when attempting to most accurately display the environmental conditions of this complex waterway. These variables are represented in the water parameters dataset (“YSI_Data_Begin_1.csv”) and observations in this dataset span from September 3, 2020 to November 7, 2025 (Rickard 2024).

The water surface elevation observations that acted as a secondary exploratory relationship to salinity were recorded at seven locations throughout water bodies of NCOS. One such location lies within our region of interest and a pressure transducer logger (YSI EXO1) is deployed on the north side of the water under Venoco Bridge. At a deployment elevation of 2.96 ft. NAVD88 (North American Vertical 50 Datum 1988), the levellogger takes water level measurements every 15 minutes. This ever-changing measurement allows researchers to contextualize changes within water composition and or concentration of solutes/particles?, providing for an increase in understanding of this complex system. This variable is represented in the metadata dataset (“NCOS_YSI_Water_Quality_Monitoring_0.csv”) and observations in this dataset span from September 3, 2020 to November 7, 2025 (Rickard 2024).

Precipitation observations, our main explanatory variable of interest within this analysis, were taken daily from the NOAA Santa Barbara 11 W, CA US station (Station ID: USW00053152) located just west of the Devereaux Lagoon (coordinates: $34^{\circ}24'50.8''\text{N}$ $119^{\circ}52'46.6''\text{W}$). Rainfall information provides our analysis with a necessary external, environmental variable in our attempt to understand the drivers of salinity change. This variable is represented in the weather dataset (“NOAA-weather-data.csv”) and observations in this dataset span from September 1, 2020 to November 30, 2025 (**NCEISB11W?**).

To characterize the structure and coverage of the dataset prior to inferential analysis, several descriptive visualizations were produced. Mean salinity values were summarized across the three monitoring sites, Venoco Bridge, East Channel, and Phelps Bridge, using boxplots to illustrate site-level variability of distribution. Observation counts were summarized by site and by elevation code using bar graphs to assess the variable coverage of available data across the sampling framework. Given the potential for non-normality in the salinity, water temperature, water surface elevation, and precipitation data, compounded by generally noted equipment malfunctions throughout the monitoring period (Rickard 2024), Spearman’s rank correlation tests were selected as the appropriate non-parametric measure of association. Spearman’s correlation was applied to evaluate the monotonic relationships between salinity

and each of these three environmental variables: water temperature, water surface elevation, and precipitation. A significance threshold of $\alpha = 0.05$ was applied to all tests.

Results

Data Characterization

Observation counts varied across the three monitoring sites, as visualized by Figure 1, with Venoco Bridge contributing the greatest number of records ($n > 600$), followed by Phelps Bridge ($n > 375$) and East Channel ($n > 300$). This imbalance in representation should be considered when interpreting site-level comparisons. Observations were further unevenly distributed across elevation codes, as seen in Figure 2, with codes 0 and 1 accounting for the majority of records and code 2 being notably underrepresented across all sites. This limited representation at deeper elevation codes constrains the reliability of any patterns observed at those levels.

Figure 1. Number of Observations at Each Site

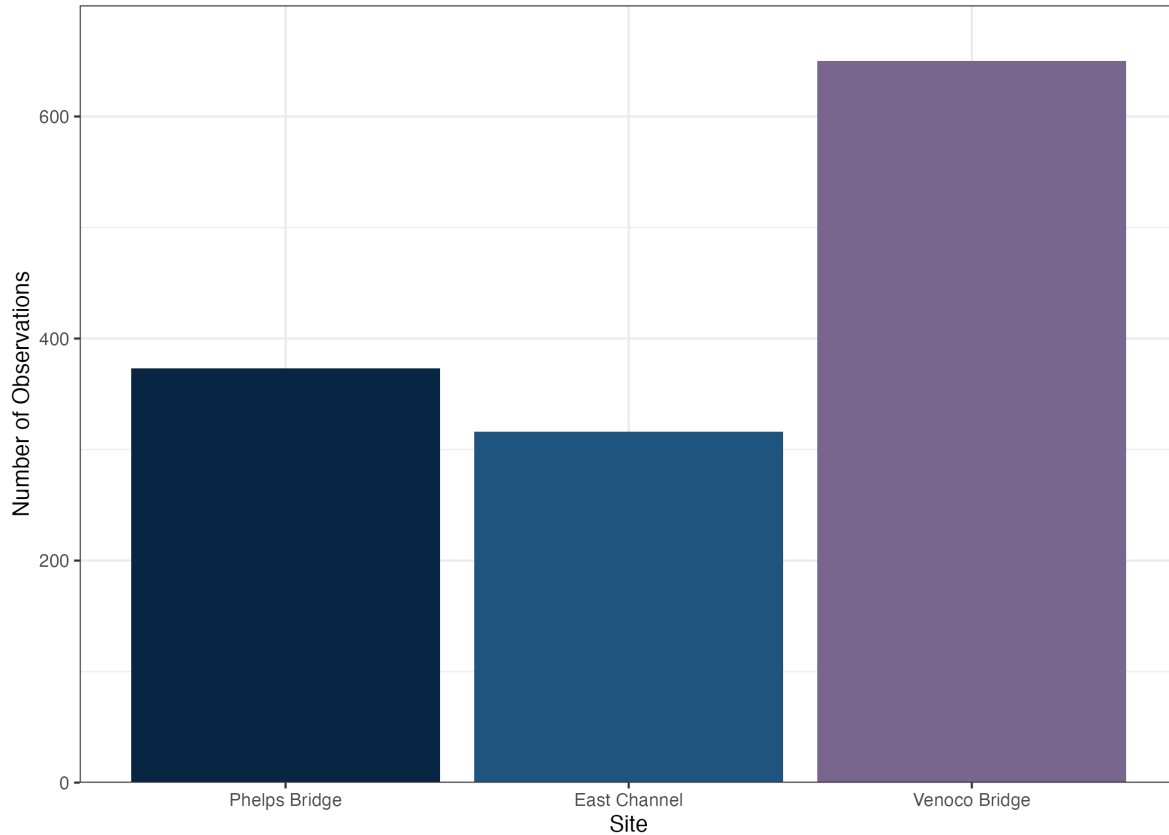


Figure 1. The number of observations taken at each of the three sites (Phelps Bridge, East Channel, and Venoco Bridge) is displayed as a bar graph. Data from Rickard (2024).

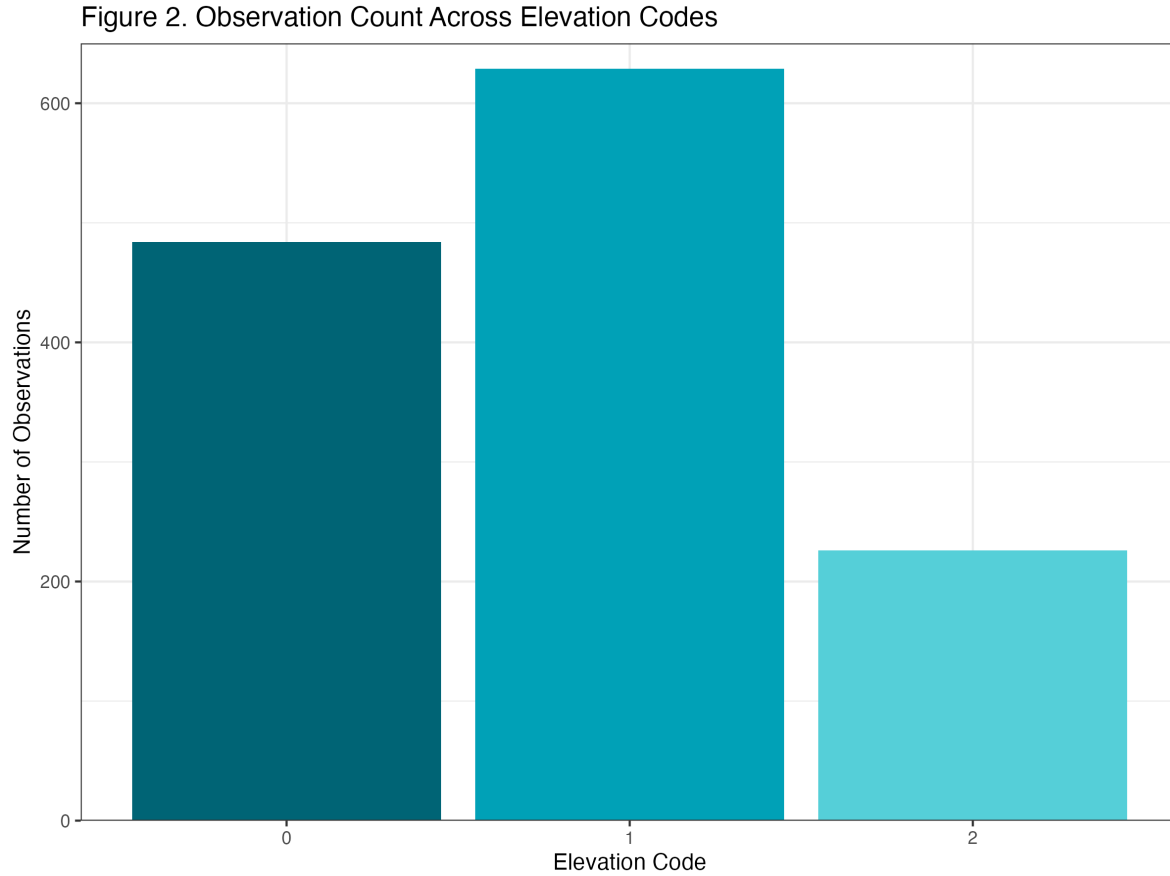


Figure 2. The number of observations taken at each elevation code (0, 1, and 2) during the study period is displayed as a bar graph. Data from Rickard (2024).

Mean salinity differed substantially across the three sites (Figure 3). Venoco Bridge exhibited the highest mean salinity at approximately 25 ppt, followed by East Channel at approximately 19 ppt, and Phelps Bridge at approximately 2.5 ppt. The interquartile ranges at Venoco Bridge and East Channel were comparably sized, indicating similar degrees of variability at those two sites, while Phelps Bridge displayed a notably compressed distribution, suggesting more consistently low and stable salinity conditions. This makes contextual sense as the Phelps Bridge site is noted to remain at freshwater levels quite consistently because of a lack of direct tidal influence (Rickard 2024).

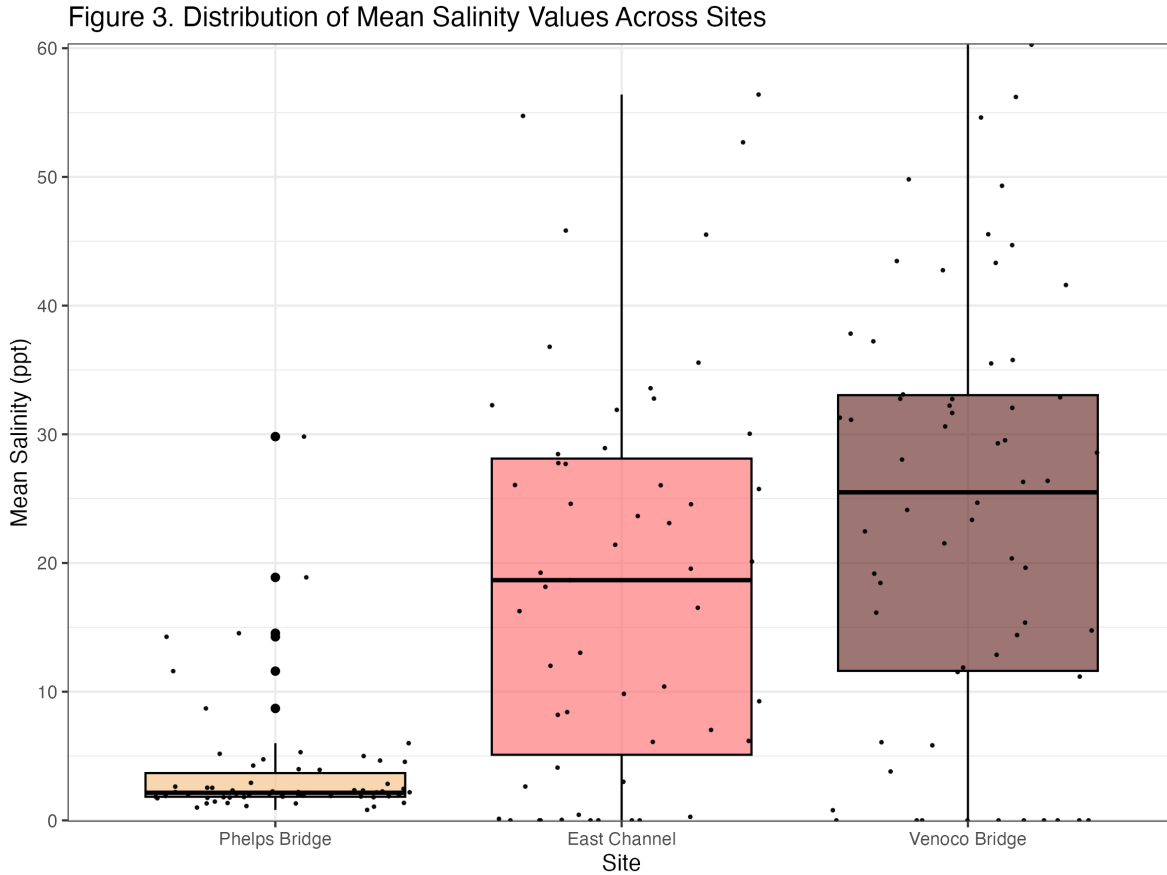


Figure 3. Mean salinity values across the study period are displayed using boxplots for each of the three sites (Phelps Bridge, East Channel, and Venoco Bridge). Data from Rickard (2024).

When salinity was examined by elevation code (Figure 4), East Channel and Venoco Bridge appeared to have lower mean salinity at elevation code 2 than at codes 0 and 1. While higher salinity at lower elevation codes might be expected given that denser saline water tends to underlie fresher water in coastal systems through saltwater intrusion dynamics (Mazhar et al. 2022), the observed inverse pattern at East Channel and Venoco Bridge is likely an artifact of the limited number of observations at elevation code 2 rather than a true reflection of salinity conditions at that elevation.

Figure 4. Mean Salinity Differences Between Elevation Codes

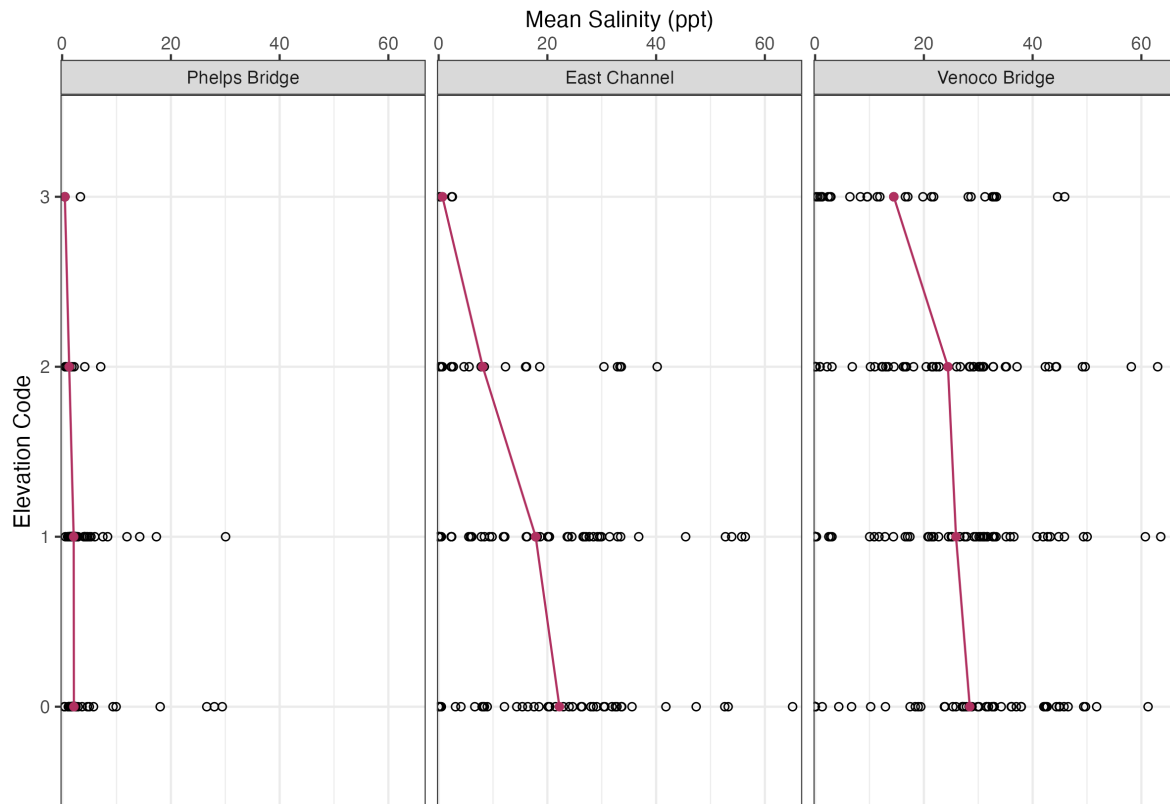


Figure 4. The mean salinity values for different elevation codes (0, 1, 2, and 3) are displayed for each of the three sites (Phelps Bridge, East Channel, and Venoco Bridge) across the study period. Data from Rickard (2024).

Precipitation and Salinity Relationship

Figure 5. Precipitation Peaks at NCOS Overlaid on Monthly Average Salinity

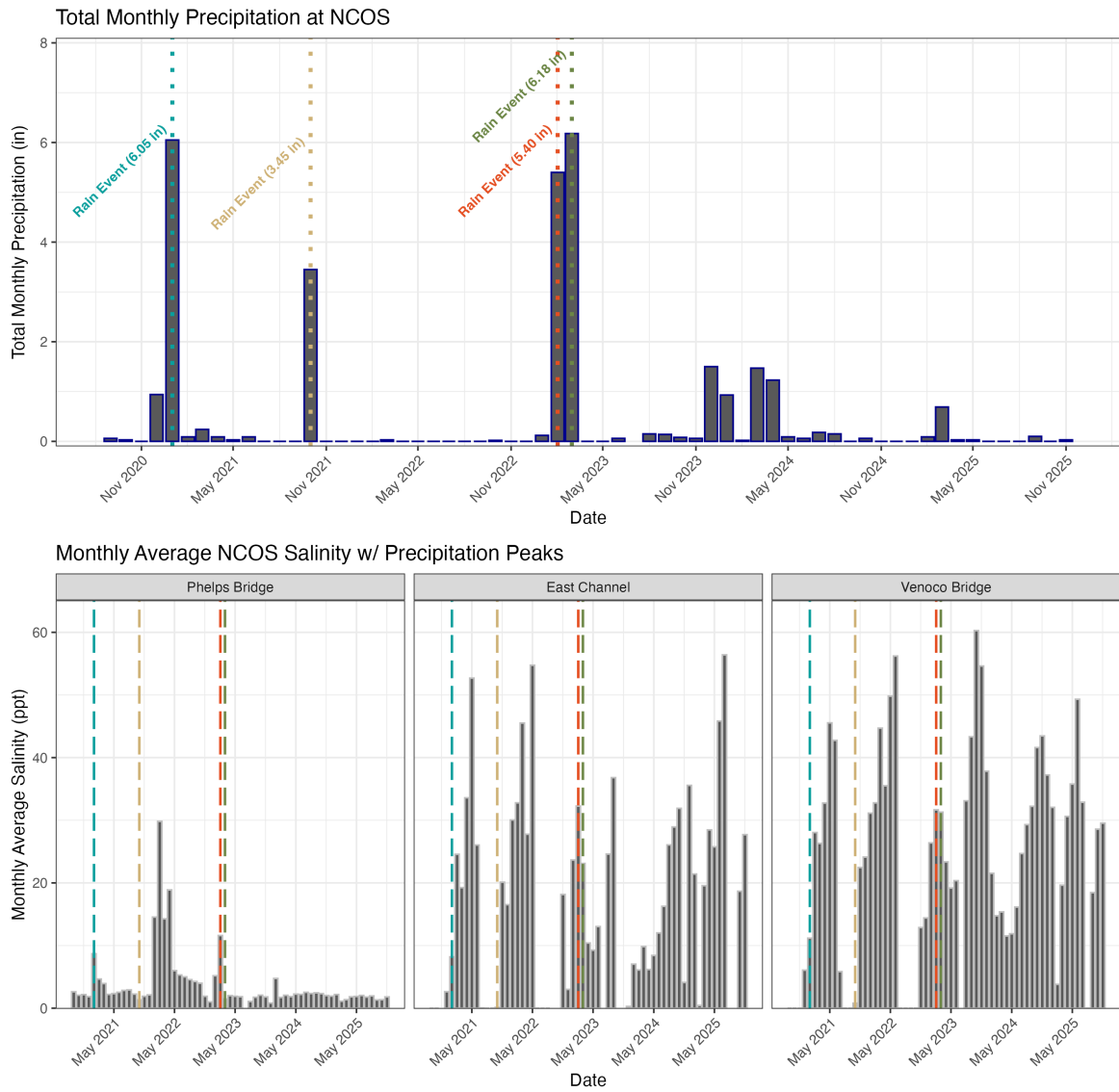


Figure 5. On the top, the total monthly precipitation is graphed by columns with the top 4 precipitation peaks depicted by colored dotted lines and labels. On the bottom, these precipitation peaks are overlaid onto the monthly average salinity for each of the three sites of interest (Phelps Bridge, East Channel, and Venoco Bridge). Data from Rickard (2024).

Monthly total precipitation across the study period was dominated by four notable peaks occurring in January 2021, October 2021, February 2023, and March 2023, with compara-

tively low rainfall recorded throughout the remainder of the record (Figure 5). When these precipitation peaks were overlaid onto monthly average salinity at each site (Figure 5), no consistent or widespread salinity response was apparent across all three sites. Isolated instances of elevated salinity coinciding with precipitation peaks were observed at East Channel and Venoco Bridge, though these patterns were not consistent enough to suggest a direct or reliable relationship as peak salinity periods were not always accompanied by a precipitation peak event. Phelps Bridge showed little discernible response to precipitation events likely because of the low recorded salinity at this particular site, making patterns difficult to observe. The relationship's scatterplot (Figure 6) displays a seemingly lacking correlation between these two variables with a heavy majority of points congregating in the low precipitation, low salinity corner (Phelps Bridge) and by the vertical pattern of points displaying that low precipitation levels can and do give rise to a large range in monthly average salinity values.

Figure 6. How Precipitation Relates to Salinity Across NCOS Sites

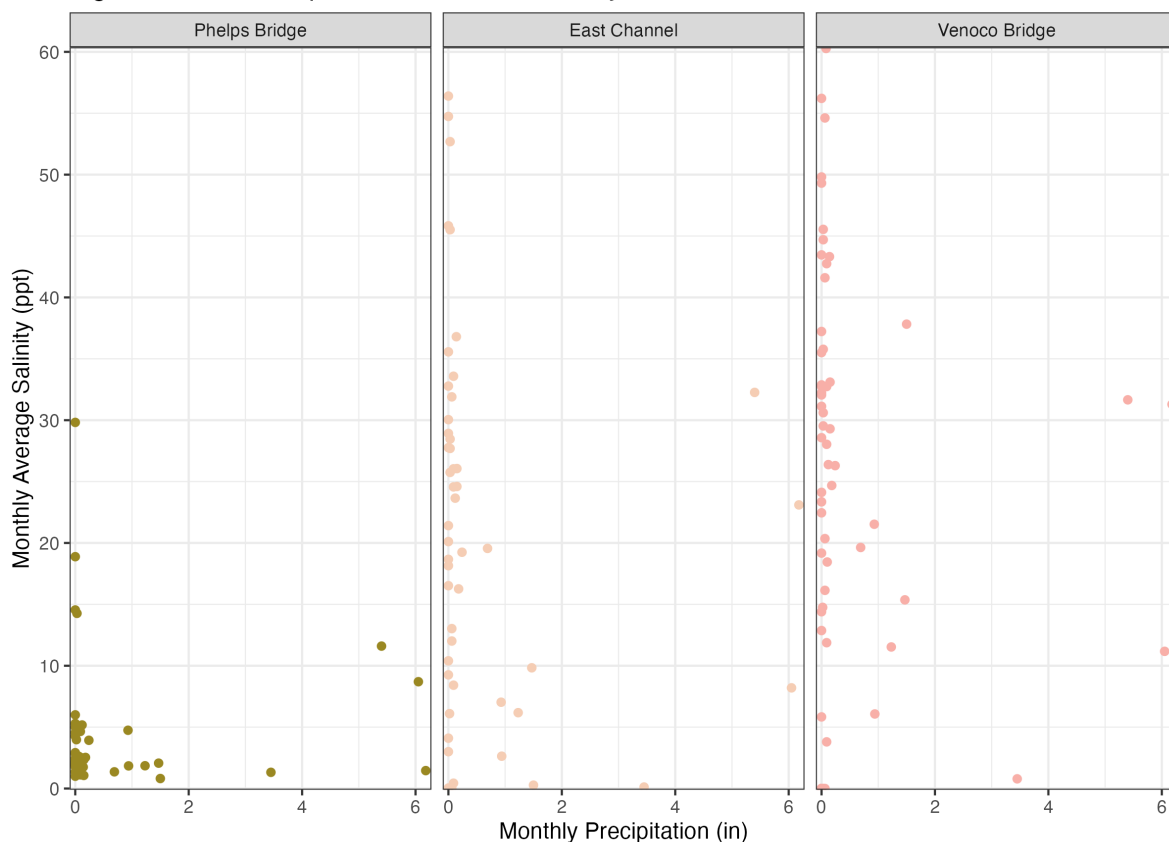


Figure 6. This figure displays the relationship between the monthly total precipitation in inches and the monthly average salinity in ppt. The figure is divided and colored by observations from each of the three sites of interest (Phelps Bridge, East Channel, and Venoco Bridge). Data from Rickard (2024).

The results of the Spearman's rank correlation test support this same observation, indicating no statistically significant relationship between monthly total precipitation and monthly average salinity ($\rho = 0.024$, $S = 932,653$, $p = 0.746$) (Figure 4). The near-zero ρ suggests negligible association between these variables at the monthly timescale, which may reflect lag effects between precipitation events and salinity response, or the dominant influence of other drivers such as tidal exchange operating independently of local precipitation inputs. The high p -value ($p = 0.746$) indicates that the observed correlation was well within the range of what would be expected by chance alone if no true relationship existed, providing no statistical evidence of an association between monthly precipitation and monthly mean salinity at this scale.

Water Surface Elevation and Salinity Relationship

Figure 7 displays the relationship between monthly average water surface elevation and monthly average salinity. While no strong or obvious linear relationship is apparent, the cluster of points exhibits a subtle positive tendency, where lower monthly average water surface elevation corresponds generally to lower monthly average salinity, and higher water surface elevations are associated with somewhat higher salinity values.

Figure 7. How Salinity Relates to Water Surface Elevation (Venoco Bridge)

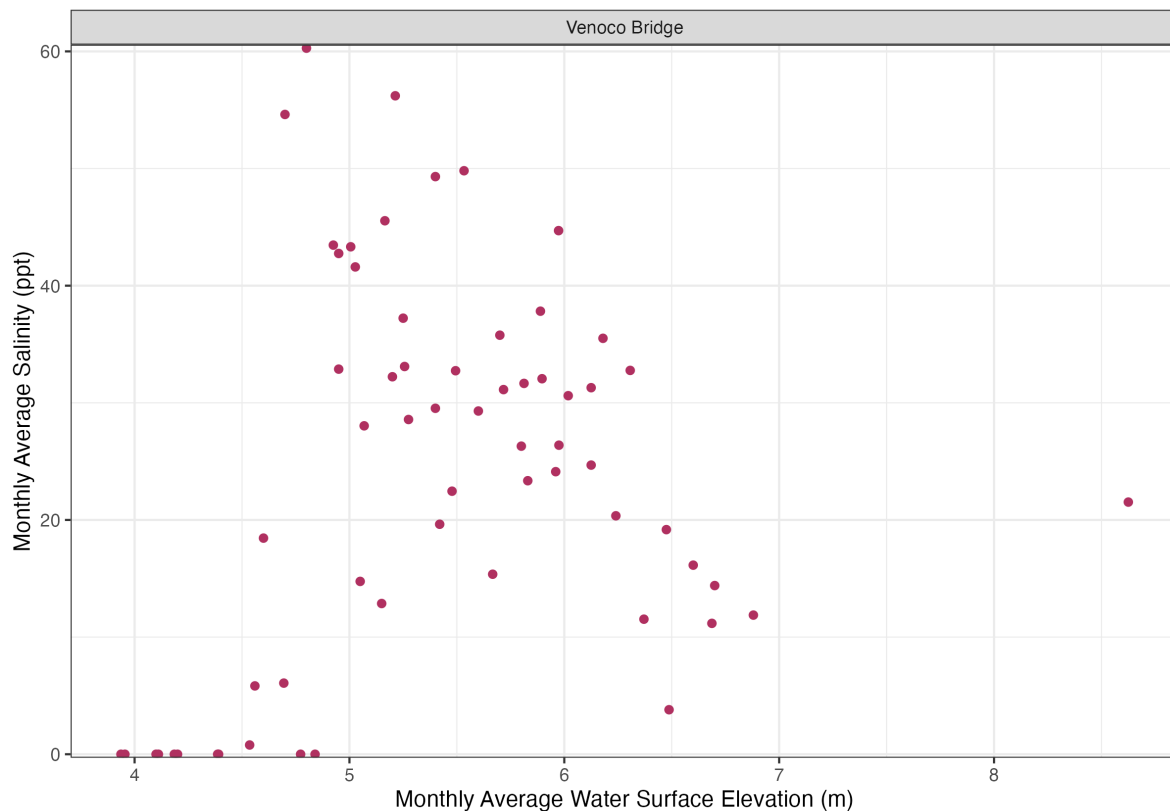


Figure 7. This figure displays the relationship between monthly average water surface elevation and monthly average salinity as maroon points. Data from Rickard (2024).

No statistically significant relationship was identified between monthly average water surface elevation and monthly average salinity ($\rho = 0.228$, $S = 30,660$, $p = 0.075$) (Figure 2). The p-value of 0.075 indicates that, assuming no true relationship existed, there would be a 7.5% probability of observing a correlation as strong as or stronger than the one obtained. While this narrowly exceeds the significance threshold of $\alpha = 0.05$, it suggests the result is not easily dismissed and may reflect a weak but potentially meaningful positive correlation between water surface elevation and salinity that warrants further investigation with a larger sample. This pattern may reflect tidal dynamics or episodic saltwater intrusion events, whereby higher water levels correspond to greater marine influence within the system, rather than the effect of increased precipitation events.

Temperature and Salinity Relationship

Figure 8 displays the relationship between monthly average water temperature and monthly average salinity across the three monitoring sites. At Phelps Bridge, the majority of points are tightly clustered at low salinity values (below 5 ppt) across the full range of observed temperatures, with only a small number of higher salinity observations scattered across the temperature range, suggesting no directional relationship between the two variables at this site. At East Channel and Venoco Bridge, salinity values are more widely distributed across the y-axis but similarly show no consistent directional pattern with increasing temperature, high and low salinity values appear across the full range of observed temperatures at both sites.

Figure 8. How Temperature Relates to Salinity Across NCOS Sites

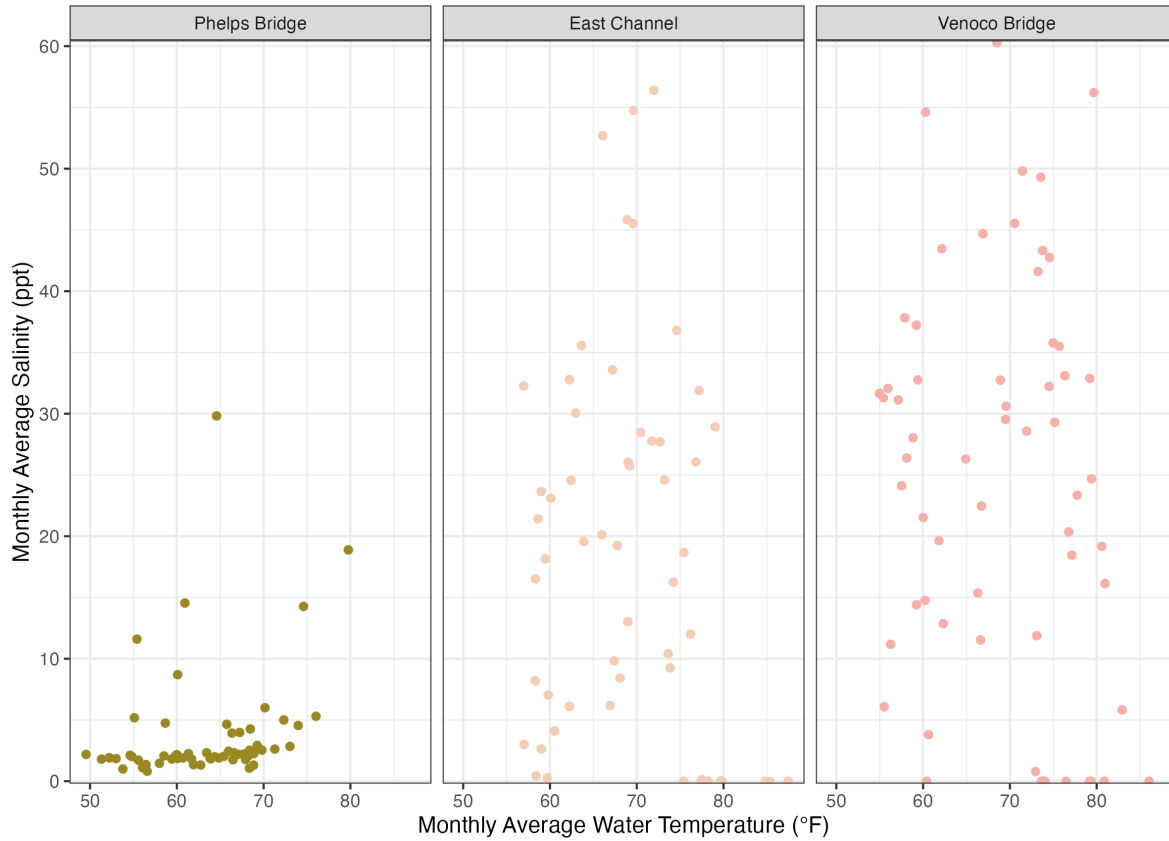


Figure 8. This figure displays the relationship between the monthly average temperature in degrees Fahrenheit and monthly average salinity for each of the three sites of interest (Phelps Bridge, East Channel, Venoco Bridge). Data from Rickard (2024).

The Spearman's correlation test confirmed no statistically significant relationship between monthly average temperature and salinity ($\rho = 0.032$, $p = 0.666$). The p-value of 0.666 indicates that the observed correlation was well within the range of what would be expected by chance alone if no true relationship existed, providing no statistical evidence of a meaningful association between these variables at the monthly temporal scale. The near-zero ρ further reinforces the negligible nature of this association, though indirect effects of temperature on salinity mediated through processes such as evaporation may not be adequately captured at this resolution.

Discussion

Salinity levels at NCOS exhibited notable spatial variation across the three monitoring sites, as illustrated in Figure 3. Venoco Bridge consistently displayed the highest mean salinity values at approximately 25 ppt, followed by East Channel at approximately 19 ppt, and Phelps Bridge at approximately 2.5 ppt. This site-level variation is consistent with the findings of Watson and Byrne (2009), who demonstrated that salinity concentrations vary across coastal wetland systems and that this spatial variability has meaningful consequences for species distribution and plant community composition. Galliari et al. (2021) further support this pattern, finding that salinity in coastal wetlands is shaped by several interacting natural processes including tidal flooding, evaporation, rainfall, and salts dissolving from sediment, and that salinity can differ substantially within a single wetland depending on elevation, groundwater flow, and tidal inundation frequency. The spatial variation observed across NCOS sites is therefore likely a reflection of differences in tidal connectivity, elevation, and proximity to marine influence across the three locations rather than the influence of any single measurable variable.

Research Question 1: How salinity relates to precipitation

As shown in Figures 1 and 6, monthly total precipitation and monthly average salinity displayed no consistent or directional relationship across the study period. The Spearman's rank correlation test confirmed this observation, returning a near-zero rho value (0.024) and a p-value of 0.746, indicating that the observed correlation was well within the range of what would be expected by chance alone if no true relationship existed. These findings are consistent with Nordio and Fagherazzi (2022), who found that while rainfall caused short-term increases in groundwater level and short-term decreases in salinity through freshwater dilution, precipitation did not exert a strong influence on salinity over longer periods such as weeks or months, the timescale of our study. Instead, salinity was more closely tied to groundwater levels, evapotranspiration, and the movement of groundwater between terrestrial and aquatic zones (Nordio and Fagherazzi 2022). This is particularly relevant to our study given that salinity was observed at a weekly timescale and further summarized at a monthly timescale, which likely obscures any short-term dilution responses to individual precipitation events. Clark (2016) further supports this interpretation in the context of NCOS specifically, noting that while rainfall lowered salinity by adding freshwater to Devereux Slough, major storm and breach events reconnected the slough to the ocean, allowing seawater to enter and raise salinity again, effectively counteracting any freshwater dilution signal at longer timescales. The relative proximity of Venoco Bridge to the slough and thus the Pacific Ocean supports the findings seen in Figure 3 that Venoco Bridge sees the highest salinity values, followed by East Channel and Phelps Bridge. The hydrology of Devereux Slough is also highly variable across years. Rickard (2024) documents that the sand berm at the mouth of the slough typically breaches once per year, temporarily making the system tidal, though in particularly wet years such as 2023 the berm breached three times due to frequent and intense storms. These breach events represent episodic, large-magnitude inputs of saline ocean water that likely have a far

greater influence on salinity than background precipitation levels, and are not captured by a monthly precipitation variable alone. Examining the relationship between mean salinity values and slough breach events, or the cumulative number of days the system remained tidal, would be a logical next step toward a deeper understanding of salinity drivers at NCOS. The sustained tidal connectivity associated with breach events likely explains salinity variability more effectively than episodic precipitation inputs, and capturing this temporal dynamic would provide a more ecologically meaningful characterization of the forces shaping salinity within the slough.

Research Question 2: How salinity relates to water surface elevation

Figure 7 displays the relationship between monthly average water surface elevation and monthly average salinity. While no strong linear pattern is apparent, the distribution of points exhibits a subtle positive tendency, where lower water surface elevations correspond generally to lower salinity values and higher elevations are associated with somewhat higher salinity. The Spearman's correlation test returned a moderately positive rho ($\rho = 0.228$, $p = 0.075$), which narrowly missed the significance threshold of $\alpha = 0.05$. While this result does not constitute statistical significance under the criteria applied in this analysis, it should not be easily dismissed. The p-value of 0.075 indicates that assuming no true relationship existed, there would be only a 7.5% probability of observing a correlation as strong as this one by chance, suggesting a weak but potentially meaningful tendency that warrants further investigation with a larger sample and perhaps a finer temporal coverage. While originally thought to be a proxy for precipitation, water surface elevation at NCOS specifically seems to be more telling of, and perhaps representative of, the occurrences of tidal breaches between Devereaux Slough and the Pacific Ocean; higher water surface elevations correspond to greater marine influence within the system and thus elevated salinity. Clark (2016) noted that salinity at NCOS often increases with depth and that breach events reconnecting the slough to the ocean were associated with marked increases in salinity, further supporting the idea that water surface elevation and tidal connectivity are meaningfully linked to salinity dynamics at this site.

Research Question 3: How salinity relates to temperature

As shown in Figure 8, there is no consistent directional relationship between monthly average water temperature and monthly average salinity was apparent across any of the three monitoring sites. At Phelps Bridge, salinity values remained tightly clustered near zero across the full range of observed temperatures, while at East Channel and Venoco Bridge, salinity values were broadly distributed across the y-axis with no clear pattern of increase or decrease with temperature. The Spearman's correlation test confirmed no statistically significant relationship between these variables ($\rho = 0.032$, $p = 0.666$), with the p-value indicating that the observed correlation was well within the range expected by chance alone under the null hypothesis of no association. While salinity is known to fluctuate seasonally in coastal wetland systems, the

Mediterranean climate of Santa Barbara limits the degree to which temperature alone drives meaningful seasonal contrasts in salinity. Anderson and Lockaby (2011) provides useful context here, finding that saltwater intrusion in tidal freshwater wetlands was most pronounced during periods of low river flow in late summer and fall, when sea levels were relatively higher and tidal connectivity was stronger. This suggests that while temperature may serve as a proxy for seasonality, the salinity changes associated with seasonal shifts at NCOS are more directly driven by tidal dynamics and the alternation between wet and dry periods than by temperature itself (Wood and Harrington 2014). The indirect pathways through which temperature might influence salinity, such as elevated evaporation rates during warmer months concentrating salts in the water column, are unlikely to be captured at the summarized monthly temporal resolution used in this analysis, and may warrant examination at finer timescales in future work.

Data Challenges and Limitations

Several challenges inherent to collecting and working with environmental monitoring data in coastal wetland systems are relevant to the interpretation of these results. The physical nature of data collection at NCOS presents practical limitations worth considering. As all three monitoring sites are located on bridges over open water, consistent sensor placement is difficult to guarantee across sampling events. Small differences in how far a sensor is lowered into the water column, or where laterally it sits relative to the current, could introduce variability in readings that is unrelated to true changes in salinity. The structure of the bridges themselves may also locally disturb water flow, meaning readings taken immediately adjacent to bridge pilings may not fully represent conditions across the broader channel or collection site. Additionally, sensors consistently utilized in estuarine environments are prone to buildup of algae and other organisms over time, which can reduce measurement accuracy between maintenance visits. Perhaps most critically, the conditions that are emerging as the most ecologically relevant to this study, large storm events and slough breaches, are also the conditions most likely to make bridge access unsafe or impractical, potentially resulting in data gaps at the very moments when salinity dynamics are most actively changing. This is further reflected in the monitoring record itself, as Rickard (2024) notes that equipment malfunction was experienced to some extent across all years of the study period, resulting in data gaps that may have influenced the representativeness of monthly averages and reduced statistical power in the correlation analyses.

The monthly temporal resolution of this dataset represents a meaningful constraint on the kinds of relationships that can be detected. As Nordio and Fagherazzi (2022) found, the effects of precipitation on salinity are most pronounced at short timescales (hours to days) through freshwater dilution of surface and groundwater. Summarizing data at the monthly scale smooths over these short-term dynamics and likely contributes to the weak and non-significant correlation observed between precipitation and salinity in this study. Daily resolution data

would better capture the salinity responses to individual storm events and breach dynamics that are known to occur at NCOS.

Beyond the variables examined in this study, several additional factors are known to influence salinity in coastal wetland systems and were not directly captured in this dataset. Tidal exchange is among the most significant drivers of salinity in systems like Devereux Slough, where the breaching of the sand berm at the mouth of the slough introduces large pulses of ocean water that can rapidly elevate salinity across the system (Clark 2016; Rickard 2024). The timing, frequency, and duration of these breach events varied considerably across the study period, from a single breach in dry years to three breach events in the exceptionally wet year of 2023, representing a source of salinity variability that a monthly precipitation variable alone cannot adequately represent. Evaporation is another important unmeasured driver, particularly during dry summer months when the concentration of salts in standing water can increase substantially in the absence of freshwater inputs (Clark 2016). Groundwater flow, which Galliari et al. (2021) identifies as a key control on salinity in coastal wetlands, also operates independently of surface precipitation and at timescales that are not easily captured by monthly monitoring data. Finally, the year-to-year hydrological variability documented by Rickard (2024), ranging from drought years with minimal rainfall and single large storm events to exceptionally wet years with nearly twice the average annual precipitation, represents a form of temporal variation that complicates the detection of stable relationships between environmental variables and salinity across the full study period.

Synthesis

Taken together, the results of this study reflect the inherent complexity of salinity dynamics in coastal wetland systems. No single environmental variable, precipitation, temperature, or water surface elevation, was found to be a statistically significant predictor of monthly average salinity at NCOS, which is consistent with the understanding that salinity in systems like Devereux Slough is shaped by the simultaneous interaction of multiple hydrological, physical, and biological processes operating across different timescales (Galliari et al. 2021). The consistent site-level differences in mean salinity observed across Venoco Bridge, East Channel, and Phelps Bridge suggest that local factors such as proximity to tidal inputs and degree of hydrological connectivity play a more dominant role in structuring salinity patterns than any single measured variable at the monthly scale. As Watson and Byrne (2009) demonstrate, this spatial variability in salinity has meaningful ecological consequences, with different salinity regimes supporting different plant communities and shifts in salinity potentially negatively affecting biodiversity (Amores et al. 2013), vegetation cover (Huckelbridge et al. 2010), carbon sequestration, and microbial activity (Zhao et al. 2017). Continued long-term monitoring at NCOS, ideally at finer temporal resolution and inclusive of tidal flux and groundwater measurements, would substantially improve the ability to isolate the relative contributions of these interacting drivers and better characterize the ecological implications of salinity variability across the site.

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